Home Credit LAP Voice Agent
Project Documentation

AI Voice Agent Assignment | Retell AI

1. Project overview

This project implements a voice-agent workflow for preliminary screening of existing Home Credit customers interested in a Loan Against Property (LAP) offer of up to ₹75 lakh. The agent's job is to gather eligibility facts, identify disqualifying conditions, route existing-loan/EMI-reduction requests away from fresh-loan screening, and provide a senior-expert handoff when the customer meets the preliminary criteria. It does not make final credit decisions.

2. Objectives

- Conduct a natural opening, confirm the customer's name, and check whether it is a convenient time to speak.
- Explain that the offer is up to ₹75 lakh and that final approval requires further assessment.
- Detect existing property-loan or EMI-reduction cases and stop fresh-loan qualification.
- Collect the required eligibility facts without repeating clearly answered questions.
- Disqualify promptly when a mandatory failure is confirmed.
- Handle requests above ₹75 lakh by asking whether the customer explicitly accepts the ₹75 lakh alternative.
- Complete the handoff only after all required facts are known and no disqualifier applies.

3. Platform and workflow

The agent is configured in Retell AI. The supplied platform screenshots document the agent overview, saved LLM simulation, and workflow. The workflow screenshot shows a call flow, post-call extraction fields (including summary, success, sentiment, outcome, preferred callback time, and collected eligibility details), a webhook node, and a save-conversation-to-memory node. The existence of a node or field does not by itself prove that downstream storage or external actions completed.

4. Conversation flow

4.1 Greeting and confirmation

Introduce the agent and company; ask the customer to confirm their name. Do not disclose financial offer details before confirmation. If the person is wrong or will not confirm, apologize without revealing the purpose and end.

4.2 Availability

Ask whether it is convenient to talk. If busy, ask for a preferred callback time, repeat it for confirmation, explain that the team will try to call then, and end. Do not claim a callback is booked unless the system confirms it.

4.3 Offer explanation

Explain the pre-approved LAP offer of up to ₹75 lakh and clarify that this is preliminary screening, not final approval.

4.4 Transfer cases

Ask whether the customer already has a property loan or wants to reduce an existing EMI. If confirmed, stop fresh-loan qualification and follow the approved transfer-specialist route without claiming an unconfigured transfer or callback.

4.5 Eligibility collection

Ask the earliest unanswered checklist item. Extract facts volunteered out of order, preserve clear corrections, and ask one question at a time.

4.6 Terminal outcomes

After a confirmed disqualifier or rejection of the ₹75 lakh alternative, give a brief explanation and goodbye, then invoke the configured end_call tool.

4.7 Qualified handoff

Only after all checks pass, explain that a senior loan expert will follow up on rates, further assessment, and next steps; final approval remains subject to assessment.

5. Eligibility rules

Field	Accepted / required condition	Failure handling
Property type	Residential, commercial, industrial	Agricultural property → disqualify
Ownership	Sole or joint	Both accepted
Original property documents	Must be available	Unavailable → disqualify
Desired loan amount	Up to ₹75 lakh	Above cap → explain cap and require explicit acceptance
Occupation	Salaried or self-employed	Collect explicitly
Income mode	Income received through a bank account	Cash income → disqualify
Estimated market value	Record estimate; no minimum specified	Do not invent a minimum
Tenure	3–15 years inclusive	Below 3 or above 15 → disqualify

Occupation and income mode are separate facts. A vague answer such as “I earn monthly” does not establish either one; the agent should clarify.

6. State-management and decision priorities

1. Extract all relevant facts from the latest response, including volunteered facts.
2. Update the known facts and apply clear corrections.
3. Check transfer conditions and confirmed disqualifiers before asking another question.
4. Handle the loan-cap alternative and wait for explicit acceptance when needed.
5. Ask only for the earliest unanswered required fact.
6. Declare preliminary qualification only when every required fact is explicit and no terminal condition applies.

7. Test coverage and results

The test archive includes seven scenarios: Qualified, Agricultural Property, Cash Income, Existing Loan, Greater than ₹75 lakh, Busy Customer, and Out-of-order. Refer to Brief_Test_Results.docx for the evidence-based status of each scenario. Some transcripts contain unintelligible audio or likely ASR errors; those cases should not be described as flawless without listening to the recording.

8. Observed limitations

- Speech recognition produced ambiguous or unintelligible turns in some calls, including the qualified and out-of-order scenarios.
- The busy-customer transcript provided in the archive ends after the agent asks for a preferred callback time; the supplied text does not prove confirmation and termination.
- A transfer-specialist statement is not proof that a transfer, callback, ticket, or CRM update occurred.
- The provided artifacts do not independently establish that post-call fields were persisted to an external system.

9. Evidence inventory

- SystemPrompt.md — final system prompt supplied separately by the project owner.
- Retell_Ai_Agent_Overview.png — agent overview screenshot.
- Voice_Agent_Platform_Workflow.png — workflow screenshot.
- Saved Simulation Test - t5.png — saved T5 simulation configuration screenshot.
- Test Scenarios/ — audio recordings, transcripts, and logs for the seven scenarios.
- Qualified-TestCase-Screenrecording.mov — additional screen recording for the qualified scenario.

10. Conclusion

The project demonstrates a structured voice-agent approach to LAP preliminary qualification, including eligibility gating, disqualification, loan-cap handling, and specialist handoff language. The supplied evidence supports successful behavior in several scenarios, while the out-of-order and busy-customer cases need careful review against their success criteria. The agent should be presented as a preliminary screening assistant, not as a loan-approval or account-servicing system.